

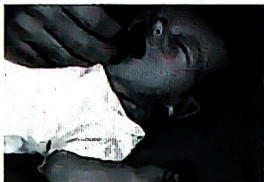
NEONATAL REFLEXES, HIE AND NEONATAL SEIZURES

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Neonatal reflexes

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- AKA immature or primitive reflex : Disappear some time after birth.

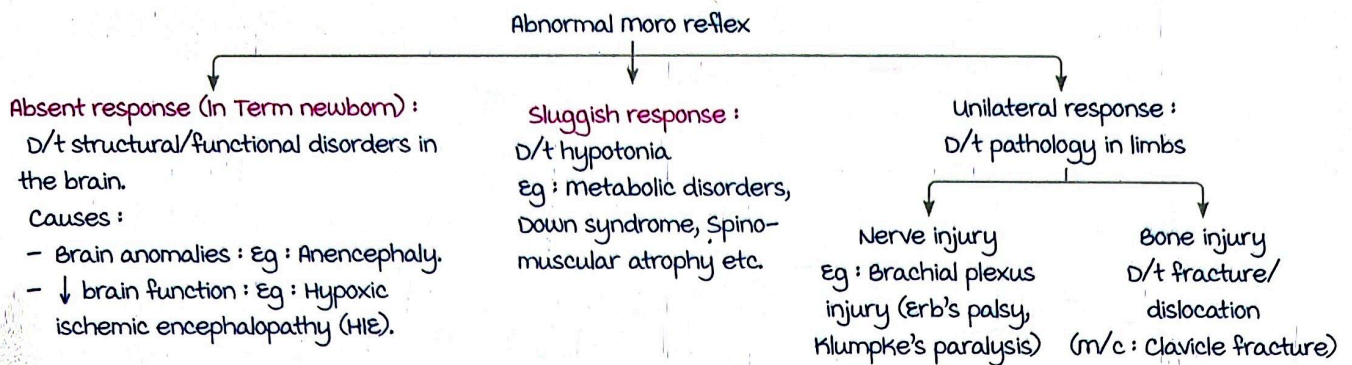
Reflex	method to elicit reflex	Onset	Fully developed by	Duration after birth
1. Palmar grasp	Involuntary palmar grasp 	28 wks of gestation	32 wks of gestation	2-3 months
2. Rooting reflex	Stimulus near mouth ↓ Face turns ipsilaterally 	28-32 wks of gestation	34-36 wks of gestation khanirfan0392@gmail.com	Less prominent after 1 month
3. Moro reflex	 Extension + Abduction of hands ↓ Flexion + adduction	28-32 wks of gestation	37 wks of gestation	5-6 months
4. Asymmetric tonic neck reflex	Neck turned to one side ↓  I/L extension C/L flexion	35 wks of gestation	1 month after birth	6-7 months

Note : Rooting reflex helps baby to find and latch to nipple during breastfeeding.

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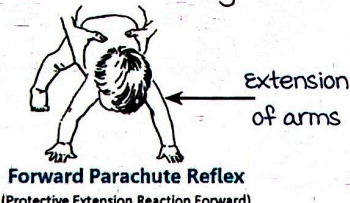
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Postnatal reflexes

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- Appear **after birth**.
- mnemonic : **SPL** (Special).

Reflex	method to elicit	Appearance (Age)	Disappearance (Age)
Symmetric tonic neck reflex	Baby held from below in prone position : • Extension of neck → B/L UL extension + LL flexion • Flexion of neck → B/L UL flexion + LL extension	4-6 months	8-12 months
Parachute reflex	Baby tilted forward (mimic falling) 	7-8 months	Does not disappear
Landau reflex	Baby held in ventral suspension in prone position : • Flexion of neck → B/L LL flexion • Extension of neck → B/L LL extension	3 months	≤9 months

Birth asphyxia and HIE Hypoxic ischemic encephalopathy

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- AKA perinatal asphyxia/**perinatal depression**.
- Impaired gas exchange → Hypoxia + Complications
 - Brain : HIE
 - multiorgan dysfunction.
 - Lactic acidosis (D/t ↑ anaerobic metabolism).

Neonatal Reflexes, HIE & Neonatal Seizures

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Definitions of birth asphyxia :

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1. WHO definition : Failure to initiate and sustain breathing after birth.

2. National Neonatology Forum (NNF) criteria :

- Based on APGAR score 1 minute after birth.
 - moderate : 4 to 6/10
 - Severe : 0 to 3/10

3. Essential criteria : American association of pediatrics (AAP) criteria : **most important.**

- Profound metabolic or mixed acidosis : umbilical cord pH < 7.
- Persistence of low APGAR score : 0-3 at 5 min.
- Signs of neonatal neurologic dysfunction : Features of HIE.
- Evidence of multiple organ involvement :
 - Eg : Kidney (Oliguria), liver (↑ liver enzymes).

HYPOXIC ISCHEMIC ENCEPHALOPATHY :

Leading cause of neonatal brain injury worldwide.

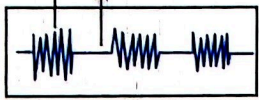
Pathogenesis :

Birth asphyxia → Hypoxia → Ischemia.

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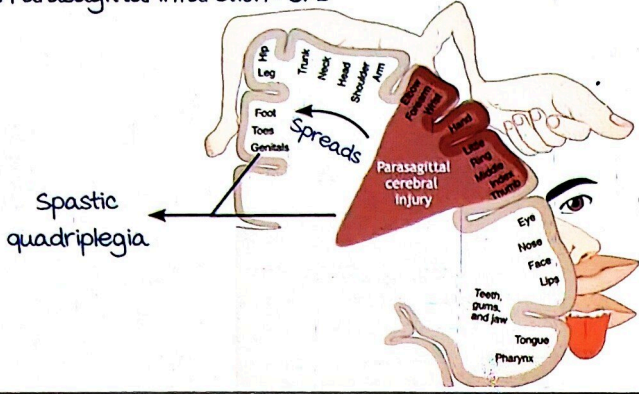
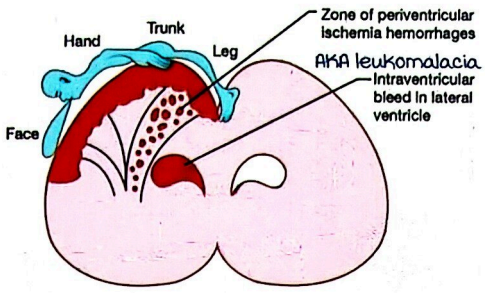
Sarnat and Sarnat staging :

- Indicates clinical features and prognosis of HIE.

	Stage 1	Stage 2	Stage 3
• Consciousness	Normal/slightly irritable	Lethargic	Comatose
• Tone	Normal	Hypotonia	Flaccidity
• Neonatal reflexes	Easily elicitable	Sluggish	Absent reflexes
• Pupil size	mydriasis (D/t sympathetic overactivity)	miosis (D/t parasympathetic overactivity)	Poor/no reaction to light
• Convulsions	Absent	Largely present	Absent
• EEG	Normal	Seizure spikes : Low voltage complexes 	Burst suppression pattern 
Outcome	99% : Normal outcome Good outcome	80% : Complete recovery 20% : complications	50% : Death 50% : Complications

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Complications**1. Cerebral palsy :**

Term babies	Pre-term babies
a. Parasagittal infarction : B/L  Spastic quadriplegia	Periventricular leukomalacia : B/L  Thinning/infarction of periventricular matter Affects motor cortex of lower limbs. Spastic diplegia
b. Focal ischemic necrosis : Hemiplegic cerebral palsy (Stroke-like condition).	
c. Selective neuronal necrosis (Eg : Basal ganglia) ↓ Dystonic/extrapyramidal cerebral palsy ↓ Extrapyramidal manifestations : Chorea, athetosis.	

2. Intellectual disability and cognitive dysfunction.**Investigations :** Diffusion weighted imaging (DWI) MRI : IOC.**management :****1. Supportive management :****a. maintain :**

- Normal temperature.
- Stable vitals.
- Blood glucose level between 75-100mg/dL (Neuroprotective range).

b. Treat seizures.

- Initial drug : Phenobarbitone.

2. Therapeutic (Induced) hypothermia :

- Gaining popularity as an established treatment modality d/t improved outcome.
- Mechanism :

Induced hypothermia → ↓ metabolism (Esp. brain) → ↓ free radicals generation → Prevents further brain damage

- method : Cooling crystals used
 - Selective head cooling
 - whole body cooling (better method)
- Temperature range : 33.5°C to 34.5°C.

Neonatal Reflexes, HIE & Neonatal Seizures

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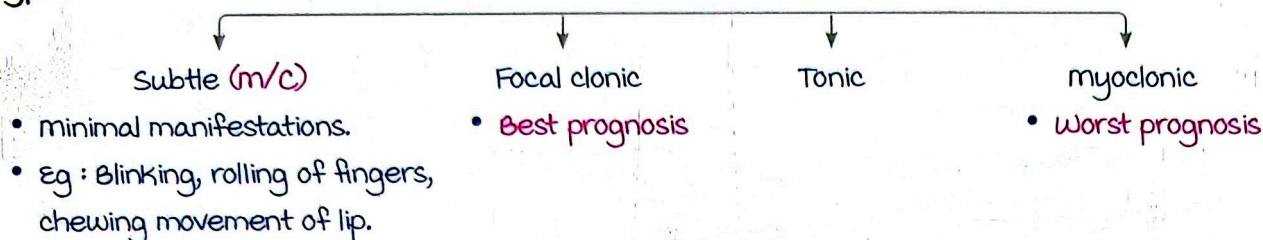
- Start : within 6 hours after birth.
- Indication : Stage 2/ Stage 3 HIE who are > 36 weeks.
- Termination : ≤ 72 hours after birth.
- C/I : Preterm < 36 wks (↑ morbidity).

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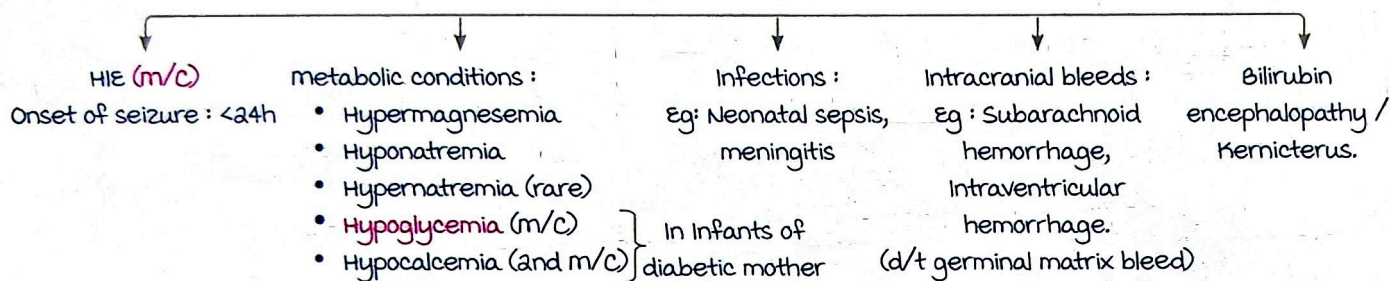
Neonatal seizures

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Types :



etiology :



management :

Stabilize the baby (Vitals & Airway, Breathing, Circulation : ABC)

Identify and correct reversible causes

Eg : Hypoglycemia (< 45 mg/dL) → IV 2ml/kg of 10% dextrose followed by infusion

if seizures continue

Anti-epileptic medication :

IV Phenobarbitone

if no response

IV Phenytoin or Levetiracetam

if refractory

IV infusion midazolam

Im Pyridoxine (Therapeutic trial) : ↑ GABA levels

- Pyridoxine is involved in glutamate metabolism.

Glutamate acid decarboxylase

+
Pyridoxal phosphate (co-factor)

Glutamate → ↑ GABA (Inhibitory neurotransmitter).